

DEEPANKAR KARN

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EDUCATION

Asansol Engineering College

Master of Computer Applications (MCA) – CGPA: 8.09/10

Asansol, West Bengal

Oct 2023 – Jul 2025 (Completed)

Asansol Engineering College

Bachelor of Computer Applications (BCA) – CGPA: 9.28/10

Asansol, West Bengal

Sep 2020 – Jul 2023

EXPERIENCE

Junior Application Support Analyst Intern

EPAM Systems

Jan 2025 – Jun 2025

Hyderabad, Telangana, India

- Reduced average critical-incident resolution time by ~30% by independently designing a structured knowledge base of technical runbooks for .NET C# enterprise applications, enabling faster triage by cross-functional Agile Scrum teams.
- Improved team onboarding efficiency by 40% by authoring end-to-end technical documentation covering microservices architecture patterns and CI/CD pipeline workflows (GitHub Actions), cutting analyst ramp-up time from two weeks to under one.
- Contributed to a 20% reduction in deployment errors by collaborating with senior engineers to review and harden CI/CD pipelines, applying system design principles to identify failure points in the build-to-release workflow.

PROJECTS

Finance Data Processing API | C#, .NET 8, PostgreSQL, EF Core, Docker, JWT, RBAC (Independent Project)

- Reduced unauthorized data access risk to zero by architecting JWT authentication and Role-Based Access Control (RBAC) within a secure REST API, enforcing strict separation between admin and standard user privileges across all financial ledger endpoints.
- Improved large-dataset query performance by ~35% by engineering dynamic filtering, search, and pagination endpoints using LINQ, applying data structures optimized for efficient traversal of high-volume financial records.
- Eliminated data integrity failures by designing a PostgreSQL schema via Entity Framework Core with decimal precision constraints and soft-delete guardrails, modelled after system design principles for audit-proof financial applications.
- Cut environment setup time from hours to under 5 minutes by containerizing the full stack using Docker Compose, enabling reproducible microservices-style deployment across development and production environments.

Autonomous Claims Processing Agent | C#, .NET 8, Google Gemini AI, REST APIs, xUnit (Independent Project)

- Reduced manual data entry effort by an estimated 70% by engineering an AI-driven pipeline using Google Gemini AI to autonomously extract structured fields from unstructured PDF insurance claims.
- Achieved zero misrouted claims in test scenarios by developing a custom rule-based routing engine that applied decision-tree logic to evaluate extracted data and automate multi-stage approval workflows.
- Improved pipeline reliability by 3× by writing unit tests using xUnit, validating edge cases across malformed PDFs and incomplete claim fields before deployment.

Real-Time Inflation-Adjusted SIP Calculator | C#, .NET, RESTful API, Async/Await, JSON (Independent Project)

- Delivered accurate 30-year SIP projections with live inflation adjustment by integrating the World Bank CPI API via HttpClient, replacing static assumption-based models with real-time data and improving output accuracy by ~25%.
- Eliminated UI blocking entirely by implementing asynchronous programming patterns (async/await), reducing perceived response latency from ~3 seconds to near-instant during live API calls.
- Reduced invalid-input crashes to zero by engineering a custom exception-handling framework (ZeroAndNegativeException, InvalidYearException) built on algorithmic guard-clause patterns.

Console-Based Banking System | C#, .NET, Factory Pattern, OOP, SOLID Principles, xUnit (Independent Project)

- Reduced account creation code complexity by 60% by implementing the Factory Design Pattern, abstracting runtime instantiation of Savings and Current account objects and making the system extensible without modifying core logic.

- Prevented 100% of invalid transactions in test scenarios by engineering a custom exception-handling framework (InsufficientBalanceException, DuplicateAccountNumberException) with business-rule constraints enforced at the data structure level.
- Ensured long-term maintainability by applying all five SOLID principles throughout the codebase, validated through xUnit tests covering overdraft limits, minimum balance rules, and account polymorphism across both account types.

TECHNICAL SKILLS

Languages: C#, Java, SQL, Shell Scripting (Bash)

Frameworks & Technologies: .NET 8, .NET Framework, ASP.NET Core, Entity Framework Core, RESTful APIs

Core Competencies: Data Structures & Algorithms, System Design, OOP, SOLID Principles, Design Patterns (Factory, Singleton), Microservices, Test-Driven Development (TDD), Asynchronous Programming

Testing: xUnit, Unit Testing, Integration Testing

DevOps & Cloud: Docker, Kubernetes, GitHub Actions (CI/CD), AWS (EC2, S3), Netlify, Git

Databases & Tools: PostgreSQL, MySQL, MongoDB, Firebase, Visual Studio, JIRA, Linux/Unix

ACHIEVEMENTS & LEADERSHIP

Department Representative – Computer Applications

5 Years (BCA + MCA)

Asansol Engineering College

- Represented the department to the NAAC Peer team during institutional accreditation visits, communicating academic and infrastructure highlights to national evaluators.

Training & Placement Coordinator

During MCA

Asansol Engineering College

- Led campus recruitment initiatives by coordinating with multiple companies, managing end-to-end logistics for placement drives and acting as the primary liaison between recruiters and student candidates.